

AD 502 – Machine Learning

Pre-Reading Material – Lecture 27

Expectation Maximization Algorithm

Introduction

The Expectation-Maximization (EM) algorithm is an iterative technique used to estimate model parameters when data contains hidden or unknown information.

It is commonly used with Gaussian Mixture Models (GMM) to identify clusters in a dataset.

What is the EM Algorithm?

The EM algorithm works by repeatedly estimating the probability of data points belonging to different clusters and updating the characteristics of those clusters.

It consists of two main steps:

E-Step (Expectation Step): The algorithm estimates the probability that each data point belongs to each cluster.

M-Step (Maximization Step): The algorithm updates the cluster parameters using the estimated probabilities.

How Does EM Work?

1. Initialization: Start with initial estimates of the clusters.
2. Expectation: Estimate the membership probabilities of each data point.
3. Maximization: Update the cluster characteristics based on these probabilities.
4. Repeat: Continue the E-step and M-step until the model converges or reaches a stopping condition.

Basic Process:

Initialize → E-Step → M-Step → Repeat → Final Model



Example

Suppose a college wants to group students based on their study habits, but the actual group membership is unknown.

- E-Step: Estimate the probability that each student belongs to a particular group.
- M-Step: Update the characteristics of each group based on the estimated memberships.
- Repeat: Continue until the groups become stable.

Applications

- Gaussian Mixture Models (GMM)
- Image segmentation
- Missing-data estimation
- Speech recognition
- Pattern recognition
- Statistical modelling

Key Takeaways

- EM is an iterative parameter-estimation algorithm.
- It alternates between the Expectation step and Maximization step.
- The E-step estimates hidden information or membership probabilities.
- The M-step updates model parameters.
- EM is commonly used to train GMMs.
- Its results can depend on initialization, and it may converge to a local optimum rather than the global optimum.

